

IJAS A H

DATA SCIENTIST

Cochin, Kerala • ijas.sayone@gmail.com • +91 7034511842 • [LinkedIn](#)

Results-driven Data Scientist and Generative AI Engineer with over 5+ years of experience in building data-driven solutions and deploying large-scale AI applications. Expertise in Python programming, predictive modeling, real-time data processing, and scalable cloud-based solutions. Proven track record of leading end-to-end AI projects that significantly enhance business productivity and efficiency.

SKILLS

Programming Languages (Primary): Python

Programming Languages (occasional use): C, C++, R, SQL, java

Cloud Platforms: Google Cloud (GCP, Vertex AI), IBM Watson, Azure, Databricks

Google cloud platform (GCP): Agent builder, Gemini models, google bucket, google logger (monitoring), cloud sql, pub/sub, App engine, Google colab workspace, Cloud source repositories, Cloud scheduler, Cloud build, Cloud run.

Generative AI: LLM fine-tuning, LLM Evaluation, Prompt Engineering, OpenAI API, Google PaLM, LLAMA 2, LangChain, GPT2, LLAMA 3. Agentic AI development, Multi Agent collaboration tool development, Langflow, Agno / Phidata, RAG, Agentic RAG, Crew AI, Hands-on Gen AI experience with fine-tuning open-source models (LLaMA, Mistral), RAG (Retrieval-Augmented Generation), embeddings

Libraries & Tools: Pandas, NumPy, Matplotlib, SciPy, scikit-learn, TensorFlow, Keras, Huggingface Transformers, MLFlow, FastAPI, Django, HTML

AI/ML Techniques: Deep Learning (CNN, RNN, DNN), Reinforcement Learning (RLHF), Explainable AI (XAI, Lime), ModelOps, Predictive Analytics, Machine Learning (Naive bayes, Decision trees, SVM, Random forest, KNN, K-Means, Hierarchical clustering), Object detection (RCNN, FRCNN), Transformers (BERT, GPT), Optimizers (adam, RMSprop), gradient descent, ML, DL, NLP

Data Engineering: Distributed Computing, Vector DB (Chroma, FAISS),

Containerization & Integration: Docker, REST APIs

Soft Skills: Cross-functional Team Leadership

Architecture Version Control: Git, DVC

Operating Systems: Linux, Windows, macOS

PROFESSIONAL EXPERIENCE

JohnDeere

Lead Software Engineer, AI

Bengaluru

2025july–Present

- Leading the design and implementation of an AgenticAI–orchestrated system for IT Service Management (ITSM) data, enabling intelligent automation of incident analysis, resolution workflows, and root-cause identification.
- Architecting a multi-agent framework integrating LLMs, retrieval systems, and autonomous decision modules, allowing AI agents to collaborate, reason, and act on complex IT operations data.
- Driving the transition from traditional automation to self-operating AI systems, improving response accuracy, reducing mean time to resolution (MTTR), and significantly enhancing service efficiency across enterprise-scale IT environments.
- Collaborating cross-functionally with data, DevOps, and ITSM teams to operationalize AI-driven insights and ensure seamless integration with existing enterprise ecosystems.

COGNIZANT

Senior Associate, Data Scientist | AI Engineer | Generative AI Engineer

Cochin

2022-june

- Worked across multiple sectors, including **retail and hi-tech**, applying AI techniques to solve industry challenges, **increasing operational efficiency by streamlining processes** and improving decision-making models.
- Led the development and deployment of a generative AI solution for a high-profile client, automating content creation processes and reducing task completion time by ~60%, resulting in a 5x increase in productivity and servicing over 40,000 users with 99.9% uptime on GCP.
 - Worked extensively with closed-source LLMs, including OpenAI LLMs and Google Gemini LLM, optimizing and fine tuning models for client-specific needs.
 - Leveraged Google Cloud and Vertex AI for model training, deployment, and monitoring.
- Led end to end development of POC and subsequently MVP for **GENERATIVE PRESENTATION TOOL**, a generative ai powered business pitch generator that automatically creates structured, professional presentations from domain-specific data and other relevant inputs.
 - Successfully secured client contracts through proof-of-concept (POC) demonstrations, showcasing the tool's ability to adapt to diverse business contexts and produce high-quality, impactful presentations.
 - Demonstrated a 20% reduction in time spent on manual presentation creation, enabling clients to focus more on strategy and delivery.
- Contributed in developing a highly scalable, high speed, efficient deep learning based text classifier for a top retail client that can classify retail products into 1000's of categories.

- Fine-tuned open-source LLMs (e.g., LLAMA 2, Mistral) for research specific needs, leading to a 2% increase in model accuracy.
- Collaborated with data scientists, engineers, and DevOps teams to deploy GenAI models, reducing integration time through efficient coordination and alignment of technical solutions.

SYSTEMS ENGINEER - R&D | CENTER OF EXCELLENCE (ML , DL)

Trivandrum, Bangalore

Data Scientist

2019-2022

- **Worked across multiple sectors** including finance, oil and gas, and telecom, applying AI and optimization techniques to solve industry challenges, **increasing operational efficiency by streamlining processes** and improving decision-making models.
- **Led a research project** to develop advanced defenses against adversarial attacks on neural networks, resulting in a **2x improvement** in the model's ability to resist attacks, as measured by a specialized adversarial dataset.
- **Conducted research across AI domains**, including **explainable AI**, **natural language processing (NLP)**, **computer vision**, **privacy-based AI**, and **responsible AI**, delivering practical solutions for various sectors such as **finance, oil and gas**, and **telecom**.
- **Utilized tools and frameworks** such as **TensorFlow**, **Keras**, and **Scikit-Learn** for **feature engineering**, model development.
- Guided interns and conducted comparative studies on Explainable AI and Federated Learning, contributing to research and development efforts.
- **Received multiple awards** for contributions to research, learning, and successful research outcomes, demonstrating **leadership** and **end-to-end ownership** in advancing AI research across diverse domains.

EDUCATION

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY

Kottayam, Kerala

M.TECH IN AI AND DATA SCIENCE - PART TIME

2022-2024

- Cumulative CGPA 9.9
- Thesis project on LLM explainability is in progress. Submitted 1st conference paper.
- Second paper is in progress.

SCMS SCHOOL OF ENGINEERING AND TECHNOLOGY

ERNAKULAM, KERALA

B.TECH IN COMPUTER SCIENCE

2015-2019

- Completed bachelors with 8.25 CGPA.

PROJECTS

QA ASSISTANT

GENERATIVE AI, DEEP LEARNING - VECTOR DB, RAG, LANGCHAIN

- Designed Production-grade Retrieval-Augmented Generation (RAG) tool deployed on Google Cloud Platform (GCP), designed to provide highly accurate and context-aware responses for user queries.
- Utilized embedding techniques with vector databases to efficiently index and retrieve relevant information, significantly improving the precision and speed of responses. Integrated LLM models , Google Gemini , to ensure nuanced and high-quality natural language understanding and generation.
- Implemented multi-agent concepts, allowing various agents to handle different aspects of information retrieval and synthesis, ensuring more efficient and specialized processing of complex queries.
- Achieved 5x productivity improvement for client's by delivering tailored insights through seamless LLM integration with the RAG architecture. Ensured scalability and reliability through cloud-native services on GCP, with automated scaling and monitoring, maintaining 99.9% uptime.

GENERATIVE PRESENTATION TOOL

GENERATIVE AI, DEEP LEARNING - VECTOR DB, RAG, LANGCHAIN, AGENTIC FLOW

- AI-powered business pitch generator that uses Generative AI to automatically create structured, professional presentations from domain-specific data and other relevant inputs. The tool analyzes industry-specific datasets, market trends, and client information to craft tailored pitch decks that align with business needs and objectives.
- Integrated high quality customized and fine tuned LLM models to ensure nuanced and high-quality natural language understanding and generation.
- The Generative AI engine seamlessly integrates with various data sources, extracting the most relevant information to ensure that each presentation is customized to the client's industry and unique requirements.
- Successfully secured client contracts through proof-of-concept (POC) demonstrations, showcasing the tool's ability to adapt to diverse business contexts and produce high-quality, impactful presentations.
- Demonstrated a 20% reduction in time spent on manual presentation creation, enabling clients to focus more on strategy and delivery.

INSURE AWARE

MACHINE LEARNING

- Built a digital solution suite for the life insurance industry, leveraging predictive modeling to generate actionable marketing insights and predict user purchasing intent.

CAR DAMAGE EVALUATOR

COMPUTER VISION, DEEP LEARNING

- A software solution that utilizes the power of deep learning (Object detection) to identify the damaged parts, evaluate the damages and to predict the possible insurance reimbursement amount.

ADVERSARIAL DEFENSE NEURAL NET

DEEP LEARNING

- A set of complex neural network architecture that could fight and prevent adversarial attacks.

MALWARE VISUALIZATION

MACHINE LEARNING / DEEP LEARNING

- The project is to implement a static analyzer tool (based on computer vision) that could detect malware better with less computational requirements.

EVALUATION OF ROBUSTNESS OF ANDROID MALWARE DETECTOR AGAINST ADVERSARIAL EXAMPLES

MACHINE LEARNING

- To evaluate the performance of machine learning-based malware scanner against adversarial examples derived through poisoning and label flipping attacks. This was a research project and it is currently published under KES 2021 International Conference.

OTHER RELEVANT PROJECTS

- Other relevant projects and additional information listed in: [LinkedIn](#)

ADDITIONAL INFORMATION

- **Certifications:** Deep learning specialization (Coursera), Infosys Certified Global Agile Developer, Infosys Certified open chain practitioner, Infosys Certified Big Data Developer, Unix Fundamentals - Infosys Certification, Infosys Certified Data Science using R Professional, IBM Applied AI Professional Certificate, Infosys Certified Python Programmer, Infosys Certified Python Associate, Certified Ethical Hacker (Expired)
- **Achievements:** 5 star rating - yearly review(cognizant), 5 star for problem solving- Hackerrank, Tagged as a Full Stack Infrastructure Professional at Infosys, Google-Coding competition(Kick start), Scored 1575th rank in Round-H, 5 star for python programming by Hackerrank
- **Awards:** Cognizant GenAI ideathon, Learning Eklavya, INSTA AWARD (Received multiple times , for Guiding ML Research and for performance), RISE AWARD (Received multiple times, Project Excellence), IBM Advanced Quantum badge for qiskit challenge 2021 , Techzooka competition award by Infosys(for presenting AI)
- **Publications:**
 1. Ah, Ijas, P. Vinod, Akka Zemari, D. Harikrishnan, Godvin Poullose, Don Jose, Francesco Mercaldo, Fabio Martinelli, and Antonella Santone. "Vulnerability evaluation of android malware detectors against adversarial examples." *Procedia Computer Science* 192 (2021): 3320-3331.
 2. I. A. H, A. A. Jo and E. D. Raj, "Exploring Explainable AI in Large Language Models: Enhancing Transparency and Trust," 2024 11th International Conference on Advances in Computing and Communications (ICACC), Kochi, India, 2024, pp. 1-7, doi: 10.1109/ICACC63692.2024.10845370. keywords: {Ethics;Analytical models;Data analysis;Explainable AI;Large language models;Computational modeling;Decision making;Data models;Critical infrastructure;Standards;Large Language Models (LLMs);Explainable AI;Transparency;Ethical AI;Model Interpretability;Accountability;Fairness;AI Applications},